Gizem Doga Filiz

Nationality: Turkish **Date of birth:** 20/01/2002 **Gender:** Female

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ABOUT ME

https://gizemdogafiliz.github.io

EDUCATION AND TRAINING

Master of Science in Automation and Control Engineering

Politecnico di Milano [15/09/2025 – Current]

City: Milano | **Country:** Italy | **Website:** https://www.polimi.it | **Field(s) of study:** Automation and Control Engineering | **Level in EQF:** EQF level 7

Bachelor's Degree in Mechatronics Engineering

Sabancı University [01/10/2020 – 20/01/2025]

City: Istanbul | **Country:** Türkiye | **Website:** https://sabanciuniv.edu/en | **Field(s) of study:** Mechatronics Engineering | **Level in EQF:** EQF level 6

WORK EXPERIENCE

Munich Institute of Robotics and Machine Intelligence (MIRMI) Lab. - TUM – Munich, Germany

Robotics Research Intern

[05/05/2025 – 15/07/2025]

- Developed vision-augmented force-impedance control framework for autonomous surface cleaning using Franka Emika Panda robot and Intel RealSense D455 camera.
- Implemented Markov Decision Process (MDP) planner with Bayesian belief updating for intelligent surface coverage motion planning and real-time stain detection on unknown 3D surfaces.
- Designed adaptive stiffness control algorithm integrating tactile feedback, surface normal estimation, and multi-level compliance adaptation for optimal cleaning effectiveness.
- Implemented multi-threaded ROS framework integrating real-time surface normal estimation using point cloud processing for tool reorientation and complete coverage of complex 3D surfaces.

Supervisors: Prof. Sami Haddadin & Dr. Kübra Karacan
Demo video: https://youtu.be/KZKU4lOgxEE

University of Naples Federico II - PRISMA Lab. – Napoli, Italy

Address: LAB B2R of University of Naples Federico II - via Claudio 21, 80125 Napoli , Napoli (Italy)

Robotics Research Intern

[12/06/2024 – 15/09/2024]

- Designed, modeled, 3D printed, and assembled hand exoskeleton with a four-bar linkage system using SolidWorks.
- Developed hardware and Machine Learning algorithms to optimize EMG sensor-based control.
- Integrated the algorithms for real-time actuation of hand exoskeleton.

Supervisor: Prof. Fanny Ficuciello & Prof. Mohammad Gohari

Özyeğin University - Biomechatronics Lab. – Istanbul, Türkiye

Biomechatronics Research Intern

[01/07/2018 – 08/08/2018]

- Interned, focused on Upper Body Exoskeleton.
- Developed electronics and software for assistive robotic devices.
- Collaborated with team to enhance exoskeleton functionality and performance.

Saha Information Technologies – Istanbul, Türkiye

Software Test Automation Developer Intern

[01/07/2019 – 08/08/2019]

- Interned in Test Automation department.
- Engaged in developing and optimizing test scripts, utilized from Java.

PROJECTS

[15/02/2024 – 15/02/2025]

Bayesian Optimization for Object Placement

- Developed a Bayesian Optimization framework with transfer learning for sequential object placement and rearrangement on cluttered surfaces.
- A key contribution of this work is a novel generalized method for Minkowski Sum estimation, scalable to n-dimensions, which emerged from the probabilistic modeling approach and extends beyond placement optimization to broader geometric applications.

Sabancı University — Graduation Project

Supervisors: Prof. Volkan Patoğlu & Prof. Esra Erdem

Link: https://youtu.be/aqMx0bXa7bk

[15/09/2024 – 15/02/2025]

Adaptive Switch IBVS: Implementation and Validation Study (Ghasemi et al., 2020) Academic reproduction study of the Adaptive Switch IBVS controller proposed by Ghasemi et al. (2020).Key Achievements:

- Successfully reproduced the staged control approach from Ghasemi et al.'s research, implementing rotation-translation decoupling in a simulated environment
- Validated the method's capability to handle extreme orientation mismatches (up to 200° rotation) where traditional IBVS fails to converge
- Conducted comparative analysis demonstrating 27-34% convergence time improvement over conventional IBVS
- Verified system stability through Lyapunov stability analysis as outlined in the original paper

Sabancı University — Vision-Based Control

Supervisor: Prof. Mustafa Ünel

Links: https://youtu.be/-r2QnFF31to | https://github.com/gizemdogafiliz/Adaptive-Switch-ImageBasedVisualServoing

Cart-Pole Swing-Up with Learning-Based Controllers

- Developed a data-driven system identification framework for the Cart-Pole system, utilizing JAX, Diffrax, and Optax to estimate system parameters and analyze motion trajectories.
- Implemented a neural network-based controller to perform Cart-Pole swing-up and stabilization, optimizing trajectory costs and comparing performance with LQR control.
- Engineered a Mixture-of-Experts controller, integrating state-dependent switching for autonomous robotic motion control, improving stability.

Sabancı University — Deep Learning for Robot Control

Link: https://github.com/gizemdogafiliz/CartPole-SwingUp-Control

[15/09/2024 – 15/02/2025]

Kinematic and Dynamic Analysis of the 3RRP Mechanism

- Developed a full dynamic model of the 3RRP mechanism, deriving closed-form forward/inverse kinematics and validating them in MATLAB/Simulink.
- Computed the largest symmetric workspace and evaluated the kinematic Jacobian and Global Isotropy Index (GII) for performance assessment.
- Formulated equations of motion using both Kane's Method and Lagrange's Method, comparing their computational efficiency and constraint handling.
- Validated the dynamic model through perturbation simulations and end-effector trajectory analysis.

[15/09/2023 – 15/02/2024]

2-DOF Planar Elbow Manipulator: Integrated Mechatronic System Design

- Designed and optimized link geometry in SolidWorks with stress analysis ensuring FOS > 2.
- Designed a buck-boost converter in LTspice for stable motor power delivery.
- Implemented a dual-loop PID control system in MATLAB/Simulink with gravity compensation and inverse kinematics for precise joint positioning.

Link: https://github.com/gizemdogafiliz/2DOF-Planar-Elbow-Manipulator

[15/09/2023 – 15/02/2024]

Student Grade Prediction through ChatGPT Conversation Analysis with ML Techniques

- Developed machine learning models to predict student homework scores from their ChatGPT conversation logs.
- Extracted and preprocessed conversation data from HTML files, and engineered features capturing student behavior patterns such as prompt complexity, error frequency, and sentiment.
- Implemented and compared multiple models

Link: https://github.com/gizemdogafiliz/Student-Grade-Prediction-via-ChatGPT-Interactions

Björnstad Syndrome BCS1L Gene VUS Pathogenicity Classifier

- Developed a Python-based algorithm to classify Variants of Unknown Significance (VUS) in the BCS1L gene associated with Björnstad Syndrome as pathogenic or benign.
- Retrieved 1000 homologous sequences via BLASTp, performed MUSCLE alignment in MEGA11, and built a Neighbor-Joining phylogenetic tree rerooted in FigTree.
- Implemented a conservation score-based classification algorithm with three thresholds, successfully classifying 26 pathogenic and 8 benign variants out of 34 VUS positions.
- Validated results against pre-classified ClinVar variants with 80% accuracy.

Link: https://github.com/gizemdogafiliz/Bjornstad-Syndrome-BCS1L-Gene-VUS-Pathogenicity-Classifier

LANGUAGE SKILLS

Mother tongue(s): Turkish

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Robotics

MATLAB / SIMULINK / Force-Impedance Control / Vision Based Control / Robot Learning / Computer Vision / Control Systems Design / Kinematics & Dynamics of Machine / Numerical Analysis / Autolev / Arduino / LTSpice / SolidWorks

Machine Learning

Bayesian Optimization / Python

Other Software Skills

C++ / Java

Soft Skills

Independent & Continuous Learner / Proactive & Self-Motivated

HONOURS AND AWARDS

[2018] First Robotics Competition (FRC)
Innovation in Control award Team Cymurghs #7466

[2018] World Robot Olympiad (WRO)
2nd Prize — Regular Category Turkey National Competition. Also reached Quarter Final in Philippines international competition.

[2018] Road to VEX (Mechaton)
Competition 3rd Prize & Engineering Design 1st Prize

[2017] First Lego League (FLL)
2nd Prize — Qualified for National Tournament

RECOMMENDATIONS

Name: Prof. Volkan PATOĞLU | Supervisor
Sabancı University, Faculty of Engineering and Natural Sciences, Human Machine Interaction Laboratory and Cognitive Robotics Laboratory, Istanbul, Turkey
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Name: Prof. Mustafa ÜNEL | Supervisor
Sabancı University, Faculty of Engineering and Natural Sciences, Control, and Robotics (CVR) Research Laboratory, Istanbul, Turkey
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Name: Dr. Kübra KARACAN; | Supervisor
Technical University of Munich (TUM), Munich Institute of Robotics and Machine Intelligence (MIRMI) AI Robot Safety and Performance Center, Munich, Germany
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